A MODULE FINITE OVER R THROUGH A RING MAP R TO S IS FINITE OVER S

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The S-module structure induced by a ring map R to S. Used inside the counted proof at lines 725-726. Lets an R-generating set be read as an S-generating set, since scalars from R act through their images in S.

Lemma 0.1. *Let $R \to S$ be a ring map. Let M be an S-module. If M is finite as an R-module, then M is finite as an S-module.*

Proof. In fact, any R-generating set of M is also an S-generating set of M, since the R-module structure is induced by the image of R in S. □

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